

9 Equity Premium Puzzle

Data: 1889-1979

- Real return on stocks: 7.04%
- Real return on riskless bond: 1.04%
- Consumption growth $= \frac{c_{t+1} - c_t}{c_t}$. mean=0.0183, standard deviation=0.0359, correlation $(g_t, g_{t+1}) = -0.1499$

Model:

- utility function $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$.
- $\frac{c_{t+1}}{c_t}$ follows a markov chain. It can be one of $\{\lambda_1, \lambda_2, \dots\}$.

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$$p_t = E \sum_{j=t+1} \beta^{j-t} \frac{u'(c_j)}{u'(c_t)} c_j = E \sum_{j=t+1} \beta^{j-t} \left(\frac{c_j}{c_t}\right)^{-\gamma} c_j,$$

hence p_t/c_t is independent of c_t . It depends only on the growth rate of consumption in period t .

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$$p_t = E \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\gamma} (c_{t+1} + p_{t+1}),$$

If the growth rate of consumption is λ_i in period t , then denote p_t/y_t as w_i .

$$w_i = \beta \sum_{k=1}^N \pi_{ik} (\lambda_k)^{-\gamma} (\lambda_k + \lambda_k w_k)$$

We may solve the above linear system for w_i .

- If $\lambda_t = \lambda_i$, $\lambda_{t+1} = \lambda_k$, then the equity return in the model is

$$\frac{p_{t+1} + c_{t+1}}{p_t} = \frac{(w_k + 1)\lambda_k}{w_i},$$

and its average rate of return is $\sum_k \pi_{ik} \frac{(w_k + 1)\lambda_k}{w_i}$

- Riskless bond price

$$q_i = \beta \sum_k \pi_{ik} (\lambda_k)^{-\gamma} 1,$$

and its rate of return is

$$(\beta \sum_k \pi_{ik} (\lambda_k)^{-\gamma})^{-1},$$

Calibration:

- Calibration of income process. $\lambda_1 = 1 + \mu + \delta$, $\lambda_2 = 1 + \mu - \delta$, $\pi_{11} = \pi_{22} = \phi$. These imply mean= μ , standard deviation= δ , and correlation= $2\phi - 1$.
- because the transition matrix is symmetric, the model spends half of its time in state 1, and half of its time in state 2.
- The paper does not choose β or γ to complete the calibration, but argues that the reasonable range for γ is $[0, 10]$.

Result is below:

